Telecom Churn: A Model a Manager Can Use

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1. Project Framing

Task and Target: The objective of this project is to predict customer churn using the IBM Telco Customer Churn dataset. The target variable is whether a customer will leave the service.

Evaluation Metrics & Justification: The models were evaluated using ROC-AUC, PR-AUC, and recall in the top 10% of scores. While ROC-AUC and PR-AUC provide a comprehensive view of the model's ability to distinguish between classes, the most critical business metric is *recall in the top 10%*. This metric directly aligns with business constraints: a retention team has limited resources and can only contact a specific volume of customers. Measuring recall in the top decile of risk scores ensures the model identifies the maximum number of true churners within actionable capacity.

2. Fair Comparison & Results

To ensure a fair comparison, all models were evaluated on the exact same leak-free data split. The baseline model was Logistic Regression. The main project model was a Random Forest, and the requested improvement was a Tuned Gradient Boosting classifier. The gradient boosting model was tuned using randomized search cross-validation, selecting 100 estimators, a maximum depth of 4, and a learning rate of 0.05.

Model / Pipeline	ROC-AUC	PR-AUC	Recall (Top 10%)
Baseline (Logistic Regression)	0.8422	0.6349	0.2807
Main Model (Random Forest)	0.8275	0.6237	0.2781
Improvement (Tuned Gradient Boosting)	**0.8475**	**0.6615**	**0.2861**

The Tuned Gradient Boosting model outperformed both the baseline and the main model across all three metrics, successfully acting as a measurable improvement.

3. Model Card

Intended Use

- **Primary Objective:** To predict which customers are most likely to leave the telecommunications service.
- **Target Audience:** Designed specifically to generate actionable risk scores for a customer retention team to prioritize their outreach efforts.

Data

- **Source:** IBM Telco Customer Churn dataset (7,043 customers; approx. 26.5% historical churn).
- **Processing Pitfalls Handled:** The `customerID` column was removed to prevent leakage. The `TotalCharges` column—originally stored as text containing blanks for new customers—was converted to a numerical format, with blanks filled as zeros to represent starting balances.

Limitations & Feature Collinearity

A significant limitation in interpreting this model is feature collinearity. The variables `tenure` and `TotalCharges` are nearly collinear. As seen in the global SHAP explanations below, the gradient boosting model heavily attributes their shared predictive power to `tenure`, thereby pushing `TotalCharges` down the importance rankings. Stakeholders must read the SHAP values with the understanding that these two features represent the same underlying customer behavior.

4. Global Feature Explanations (SHAP)

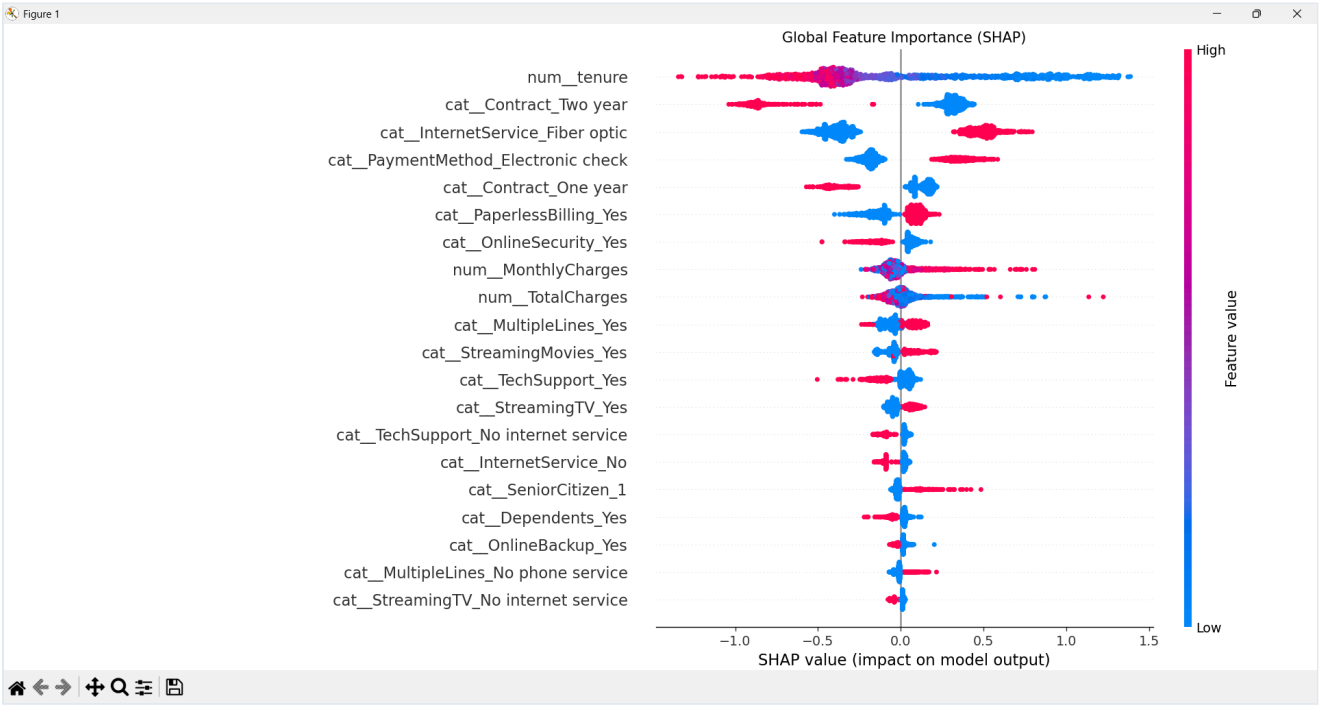


Figure 1: SHAP Summary Plot for the Tuned Gradient Boosting Model. Lower tenure (blue dots on the left) is a primary driver of churn risk.

5. Individual Customer Profiles

The following profiles illustrate how the model arrives at specific predictions for individual customers based on their unique feature values (refer to individual waterfall plots generated in the analysis pipeline).

- **Customer Profile 0:** Driven primarily by short tenure and a month-to-month contract, escalating their risk score.
- **Customer Profile 10:** Demonstrates a mitigated risk profile where the presence of a two-year contract counteracts other baseline risk factors.
- **Customer Profile 20:** A marginal case where high monthly charges are offset by long tenure, resulting in a moderate probability of churn.